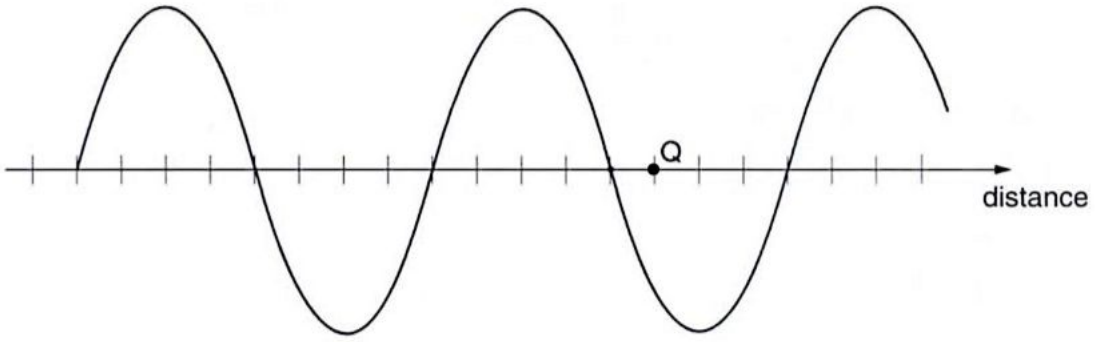
11 The diagram shows the displacement of a transverse wave along a stretched cord at a particular instant. The wave travels from left to right with frequency 20 Hz.



What is the shortest time before the displacement at Q becomes zero?

A 6.25 ms
B 18.8 ms
C 31.3 ms
D 43.8 ms

Ans: A

$$T = \frac{1}{f} = \frac{1}{20} = 50\ \text{ms}$$

From the diagram, the wavelength corresponds to 8 divisions, while Q is approximately one division beyond the zero-displacement position P. Since the wave travels from left to right, the displacement at a fixed point changes according to the part of the wave profile that is progressively arriving from the left.

The phase difference corresponding to one division is therefore

$$\Delta\phi = 2\pi\frac{1}{8} = \frac{\pi}{4}$$

$$\Delta t = \frac{\Delta\phi}{2\pi}T = \frac{1}{8}(50\ \text{ms}) = 6.25\ \text{ms}$$